

Tyler Takimoto

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EDUCATION

Bachelor of Science, Computer Engineering
University of Central Florida

May 2026
Orlando, Florida

WORK EXPERIENCE

Mitsubishi Power Americas
Engineering Associate

May 2025 – May 2026
Orlando, Florida

- Built Python tools to automate manual workflows across 3 teams, saving hours of engineering time.
- Developed a machine learning program utilizing PyTorch to automate turbine repair image processing, achieving 99% accuracy and classifying repairs 2x faster than manual reading.
- Redesigned an error-prone internal SQL database editor, resolving cross-team workflow bottlenecks and cutting usage time by 1/3.
- Introduced data logging and automated backups to repair machines, providing faster failure recovery.

PROJECT EXPERIENCE

NIMBL, Senior Design – Embedded Software Engineer

August 2025 – May 2026

- Designed a low-power pulse-oximeter using a TI MSP430 MCU focused on portability and accuracy.
- Created custom C++ libraries to interface a MAX32664 biometric sensor, CC1101 radio module, and HT1261 LCD driver, accurately measuring heart rate within 2bpm and transferring over 10 meters.
- Crafted a Python visualization app to receive the transmitted data, displaying the real-time heart rate and blood oxygenation within 10ms of transmission.
- Collaborated within a cross-function team to author a 120-page technical design report detailing our hardware choices, system architecture, and team goals.

fAlnman, Knight Hacks VIII – Project Lead

October 2025

- Directed a development team through a 36-hour hackathon, designating roles based on technical expertise to create a full-stack AI-assisted learning application.
- Built upon the Feynman studying technique by integrating an AI listener into a chat box plus drawing canvas, expanding the user's ability to teach.
- Developed the Python backend, building our APIs to connect the frontend, MySQL database, and our Google Gemini model.
- Utilized DigitalOcean to host our web service with a custom HTTPS domain, automating updates via a YAML CI/CD pipeline to auto-deploy new GitHub commits.

Pocket Professors, UCF – Frontend Developer

May 2025 – July 2025

- Contributed to a MERN full-stack UCF-based card collecting web and Android application.
- Implemented the ReactTS frontend and integrated backend APIs to enable account creation and dynamic catalogue tracking with live card searching.
- Developed the Flutter-based Android app, maintaining parity between the website and application.

Polly the Parrot, Knight Hacks VII – Software Lead

October 2024

- Built a talking AI parrot for a 36-hour hackathon, achieving 2nd place for Best Hardware Hack.
- Integrated ChatGPT and IBM Watson in the backend to build a seamless Speech-to-Text and Text-to-Speech interaction loop, creating a pirate-parrot focused on personal engagement.
- Connected our Arduino hardware in Python to sync audio and beak movements for lifelike motion.

SKILLS

- **Back-End:** Python, C, C++, Java, Bash, SQL
- **Front-End:** HTML, CSS, JavaScript, TypeScript, React, Vite, NodeJS, Flutter, Tailwind CSS
- **Software:** Linux, Docker, GitHub, YAML, MySQL, Machine Learning, Fusion360, Microsoft 365